Machine Learning in NBA Over/Under Betting

Business 35137: Machine Learning in Finance

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1. Introduction

In sports betting, a total (also called an over/under) is the sportsbook's prediction of the combined points scored by both teams in a game. Bettors wager on whether the actual combined score will be over or under this number. Standard odds of -110 mean a bettor must risk $110 to win $100, implying the sportsbook takes a commission (the vig) on every bet. To break even at -110 odds, a bettor must win at least 52.4% of bets.

The Vegas line is set by professional oddsmakers who incorporate team statistics, injuries, matchup history, public betting patterns, and other market signals. It is widely regarded as the most efficient public forecast of game outcomes. Any model that claims to beat the line must find a small, repeatable edge that the market has overlooked.

This report investigates whether machine learning models can predict NBA game totals more accurately than the Vegas line, and under what conditions selective betting strategies can generate positive returns. We train six model classes spanning linear, dimension-reduced, tree-based, and kernel-based approaches using walk-forward validation on 11 NBA seasons (2013/14 through 2023/24), then test the nominated strategy on a held-out 2024/25 season that was never used in any modeling decision.

Approach

We construct lagged, within-season team features from box score statistics to avoid lookahead bias. Six model classes are trained: Ridge, Lasso, ElasticNet, Principal Component Regression (PCR), XGBoost, and RBF+Ridge (Radial Basis Function kernel approximation combined with Ridge regression). Models are evaluated via walk-forward backtesting with strict temporal ordering, tuned on betting accuracy rather than regression error, and subjected to rigorous statistical testing including Bonferroni-corrected significance tests and McNemar's paired comparison. Selective betting with confidence thresholds is explored to concentrate bets where the model's edge is strongest. Finally, bankroll simulations with 10 sizing rules stress-test whether statistical edge translates into economic viability, and real-world frictions (slippage, market depth, account restrictions) are quantified.

2. Data

Two primary data sources cover 18 NBA seasons (2007/08 through 2024/25). Betting data contains approximately 23,000 games with the Vegas total line, point spread, and final score. NBA box score data provides team-level statistics for each game: points, field goal percentage, three-point percentage, free throw percentage, rebounds, assists, steals, blocks, turnovers, and personal fouls. Historical box scores cover 2007/08 through 2022/23; the 2023/24 and 2024/25 seasons are retrieved via the NBA Stats API and merged with the archive.

The two sources are merged on date and team matchup. Team abbreviations are standardized to account for franchise relocations (e.g., NJN to BKN, SEA to OKC). Each game is assigned a season-end year. The 2024/25 season (1,293 games) is held out entirely from model training and evaluation; the remaining 21,397 games form the walk-forward training and test pool.

Figure 1: Data Overview — 2007-08 through 2024-25 NBA Seasons

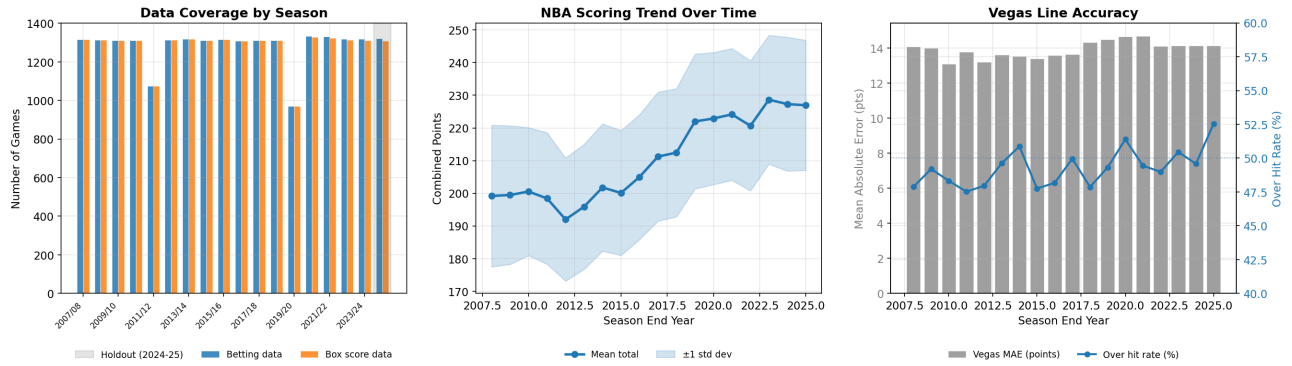

Figure 1: Data coverage by season, NBA scoring trend, and Vegas line accuracy.

3. Feature Engineering

Feature engineering transforms raw game data into 114 numerical inputs organized into seven categories: Vegas line

features (3), totals and scoring (18), shooting efficiency (24), rebounding and hustle (16), playmaking (8), schedule and rest (11), context indicators (6), and interaction features (28). All features are strictly lagged: rolling means are computed over each team's previous games within the same season using a shift(1) operation, ensuring no future information leaks into any prediction.

Rolling averages are computed at two window sizes: L5 (last 5 games) to capture short-term form, and L10 (last 10 games) for medium-term stability. For the first game of each team's season, no prior data is available, so these games are dropped. An early-season indicator variable flags the first 10 games of each team's season, where rolling statistics are based on limited observations and are therefore noisier.

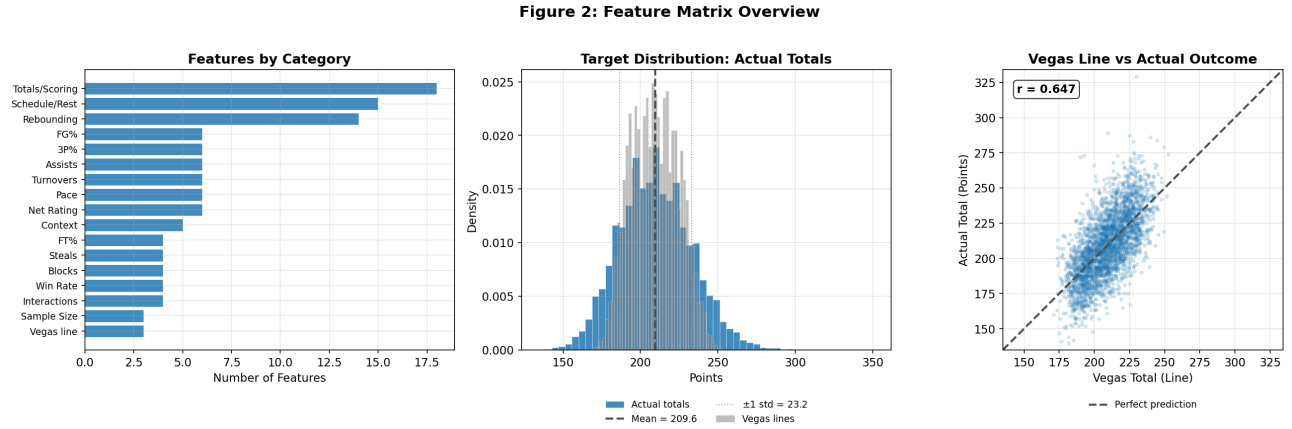

Figure 2: Feature composition, target distribution, and Vegas line vs. actual outcomes.

Univariate correlation analysis confirms that Vegas-derived features and recent scoring averages are the strongest individual predictors (r = 0.55-0.65), while schedule and rest features show near-zero marginal correlation. Interaction features (e.g., pace x pace, spread x total) are included to help linear models capture non-additive effects.

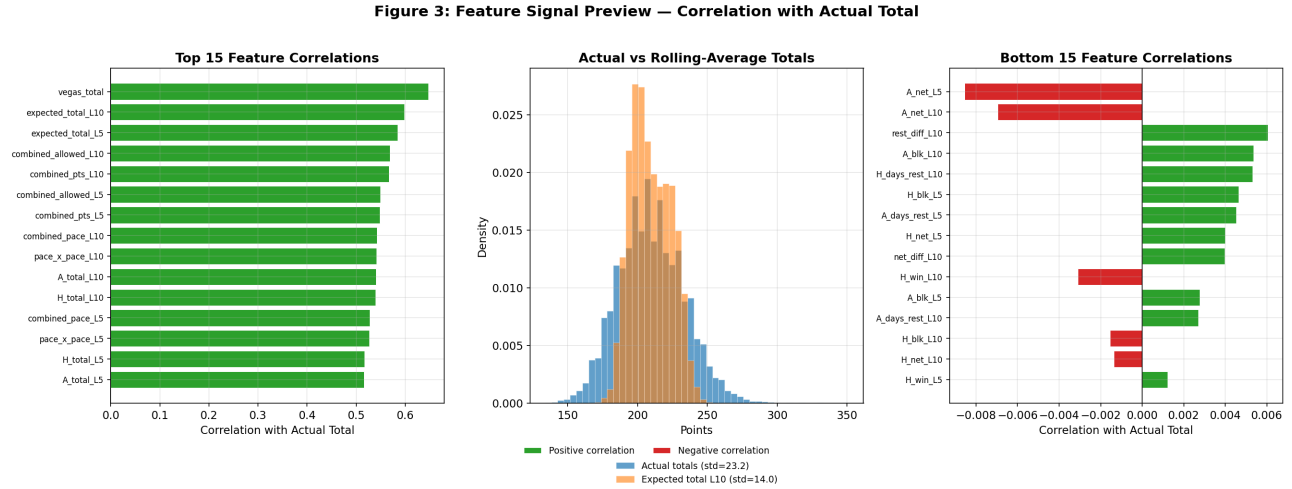

Figure 3: Top and bottom features ranked by absolute correlation with actual game totals.

4. Model Training and Walk-Forward Evaluation

Walk-forward backtesting enforces strict temporal ordering: for each of 11 test seasons (2013/14 through 2023/24), the model trains on all available data before that season and predicts the test season. The training set grows by approximately 1,300 games per iteration. Hyperparameters are tuned via 5-fold TimeSeriesSplit within the training set, optimizing betting accuracy rather than RMSE, since the downstream task is predicting over/under outcomes.

A key design choice is exponential recency weighting controlled by a half-life parameter, tuned jointly with model hyperparameters. Each half-life/hyperparameter combination is evaluated during cross-validation, and the combination that maximizes betting accuracy is selected for that season. This allows the models to adapt to the NBA's evolving pace and scoring trends without manually specifying a recency window.

Pooling predictions across all 11 test seasons yields approximately 11,000-13,000 out-of-sample bets per model. A

no-bet filter excludes games where the model's prediction is within 0.5 points of the Vegas line.

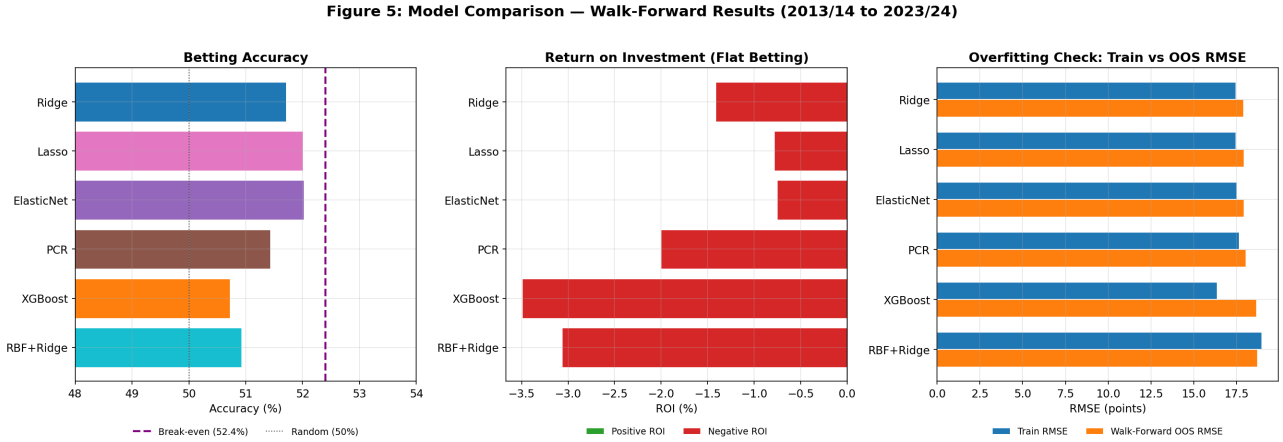

Figure 5: Walk-forward results — betting accuracy, ROI, and overfitting check (train vs. OOS RMSE) for all six models.

Key Findings

All six models cluster between 50.7% and 52.0% accuracy. The linear models (Ridge at 51.7%, Lasso and ElasticNet at 52.0%) achieve the highest accuracies, while XGBoost (50.7%) and RBF+Ridge (50.9%) trail behind. No model reaches the 52.4% break-even threshold when betting every qualifying game, so all ROI values are negative. This is the expected result in an efficient market.

Simple models outperform complex ones. The overfitting check (Figure 5, Panel 3) reveals that linear models show minimal train-vs-OOS RMSE gaps (~0.4 points), while XGBoost shows a 2.3-point gap, confirming it memorizes training patterns that do not generalize. In an efficient market where exploitable signal is small and noisy, the additional flexibility of complex models becomes a liability.

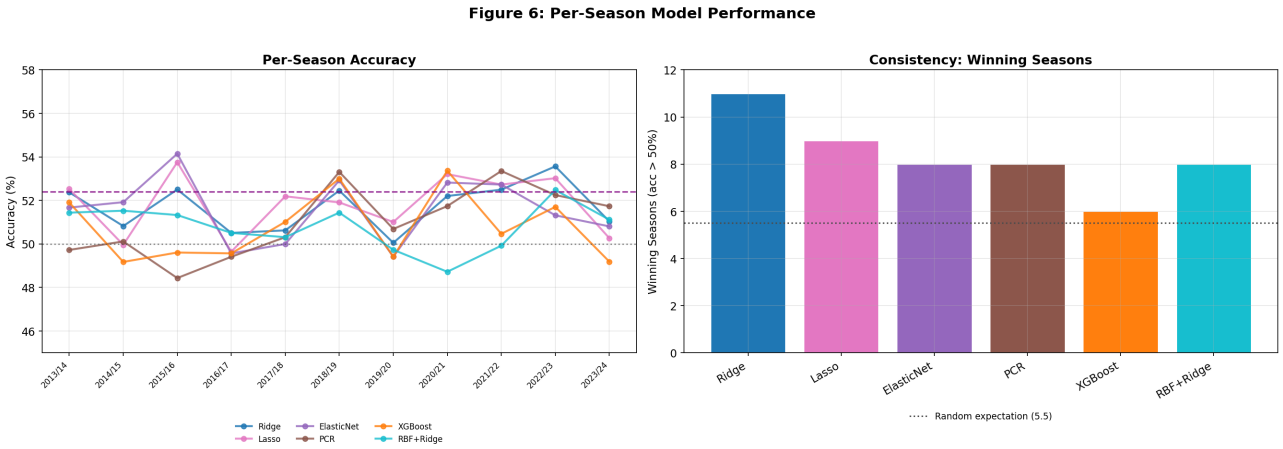

Figure 6: Per-season accuracy and consistency across 11 walk-forward test seasons.

Per-season analysis reveals that accuracy is volatile (48-55% range), with no model consistently beating 52.4% across all seasons. The linear models tend to move together, suggesting they extract similar signal. Lasso achieves the highest season-above-50% count (10 of 11 seasons), demonstrating the most consistent performance.

5. Statistical Significance Testing

With over 11,000 bets per model, even small deviations from 50% require formal significance testing. We apply four complementary tools: Wilson score confidence intervals, bootstrap resampling (standard and block bootstrap to account for temporal correlation), Bonferroni-corrected p-values for multiple comparisons, and McNemar's paired test for pairwise model comparison.

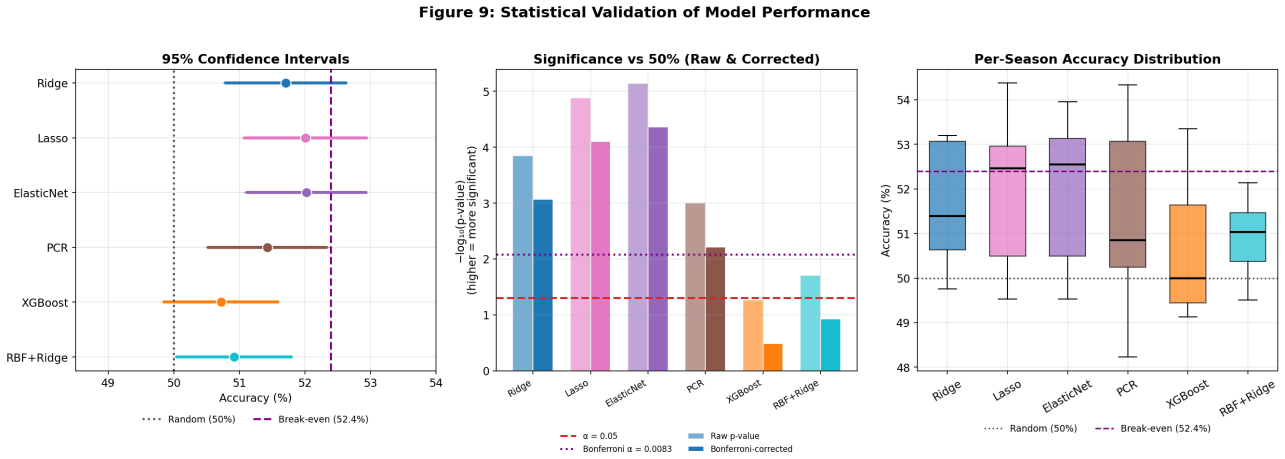

Figure 9: Confidence intervals, significance testing (raw and Bonferroni-corrected), and per-season accuracy distributions.

After Bonferroni correction for testing 6 models, four survive with significance against 50%: Ridge, Lasso, ElasticNet, and PCR (all $p < 0.006$). XGBoost ($p = 0.054$) fails to reach significance even before correction. All three CI methods (Wilson, standard bootstrap, block bootstrap) produce nearly identical intervals, confirming that temporal correlation does not materially inflate confidence. However, every model's CI includes 52.4%, meaning we cannot confirm profitability with flat betting.

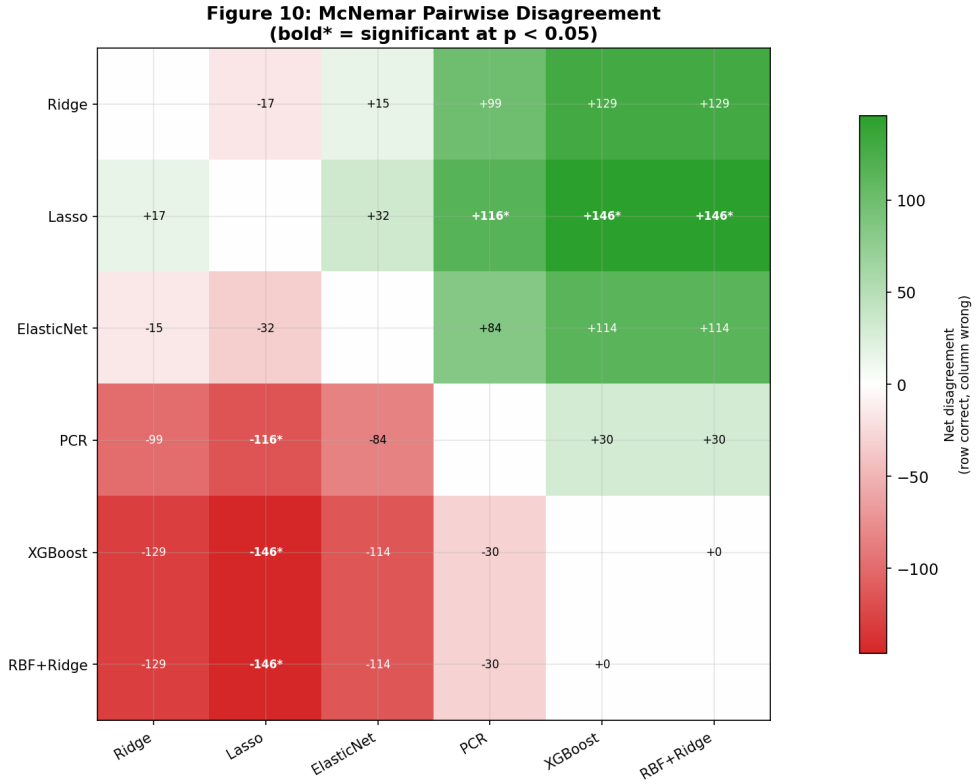

Figure 10: McNemar pairwise disagreement heatmap. Asterisks denote statistically significant differences ($p < 0.05$).

McNemar's test confirms that Lasso significantly outperforms PCR, XGBoost, and RBF+Ridge in pairwise comparisons. The linear models (Ridge, Lasso, ElasticNet) do not differ significantly from each other, consistent with extracting similar signal from the features.

6. Selective Betting with Confidence Thresholds

Since no model is profitable when betting every game, we test whether restricting bets to games where the model strongly disagrees with the Vegas line can push accuracy above break-even. The absolute difference between the model's prediction and the Vegas total serves as a confidence measure. We sweep thresholds from 0.5 to 20 points,

subject to dual validity constraints: at least 300 total bets globally, and at least 10 bets in every individual season.

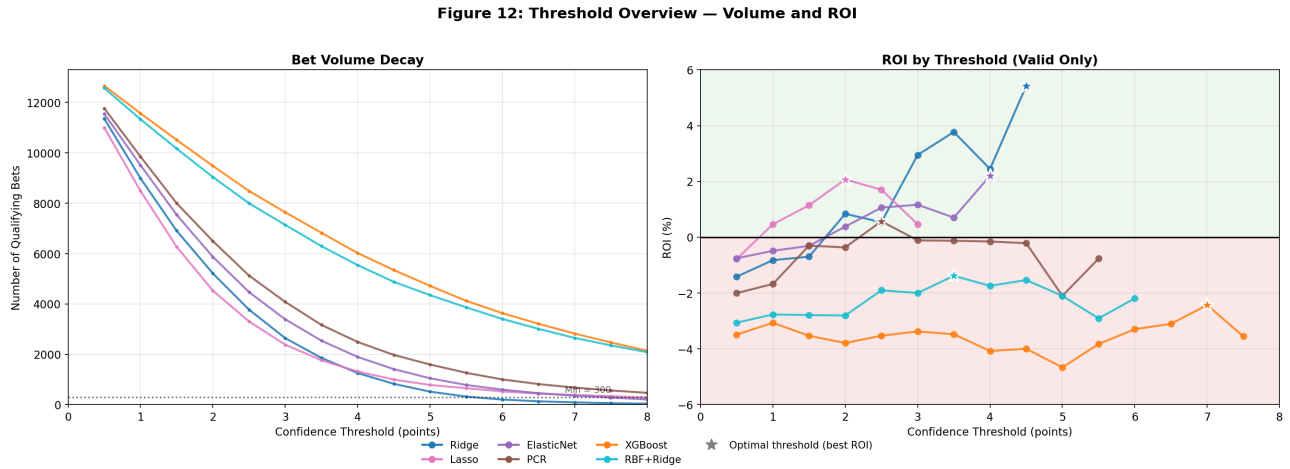

Figure 12: Bet volume decay and ROI by confidence threshold for all six models. Stars mark optimal thresholds.

Four models achieve positive ROI at valid thresholds. Ridge leads at +5.4% ROI (55.0% accuracy, 826 bets at threshold ≥ 4.5), followed by ElasticNet (+2.2%, 1,896 bets at ≥ 4.0), Lasso (+2.1%, 4,529 bets at ≥ 2.0), and PCR (+0.6%, 5,126 bets at ≥ 2.5). XGBoost and RBF+Ridge never achieve positive ROI at any valid threshold.

The explanation lies in prediction variance. Linear models have prediction standard deviations of 2.4-2.8 points, making conservative adjustments to the efficient Vegas line. When they strongly disagree with Vegas, it likely reflects genuine signal. XGBoost and RBF+Ridge have standard deviations of approximately 6 points, routinely making extreme predictions driven by overfitting rather than genuine market mispricing.

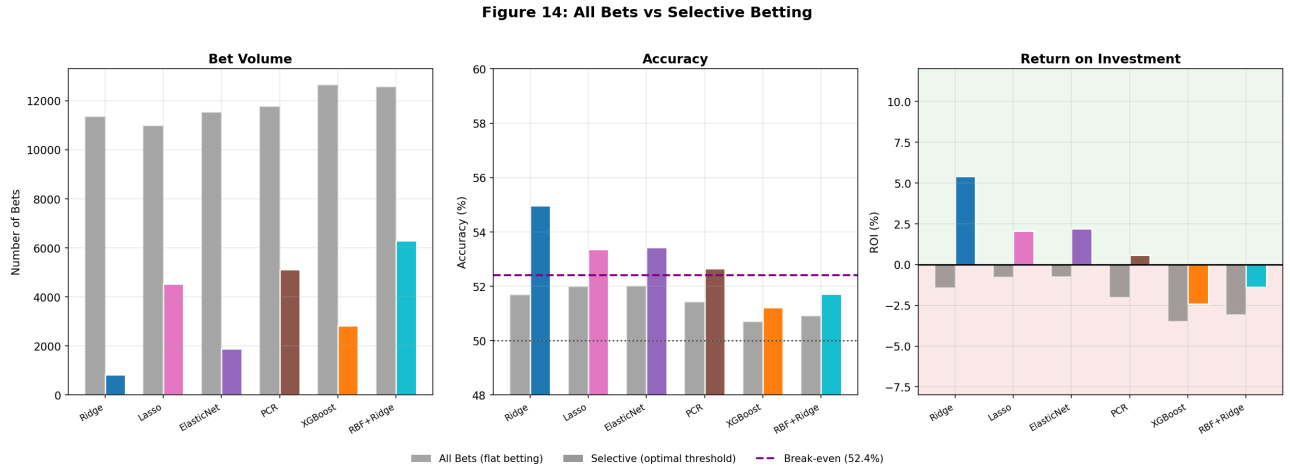

Figure 14: Comparison of base (all bets) vs. selective betting — volume, accuracy, and ROI for each model.

After Bonferroni correction, five models remain significantly above 50% at their optimal thresholds. However, no model's selective accuracy is significantly above 52.4%. The evidence for profitability is suggestive but not statistically conclusive, as the smaller sample sizes from selective betting widen the confidence intervals.

7. Bankroll Simulation and Holdout Deployment

The four qualifying models are subjected to bankroll simulations across 10 sizing rules, ranging from conservative flat dollar amounts to the theoretically optimal Kelly criterion and its fractional variants. Starting from a $10,000 bankroll, the simulations run across all 11 walk-forward seasons to evaluate growth, drawdown, and consistency.

Figure 16: Backtest Bet-Sizing Comparison (11 Walk-Forward Seasons)

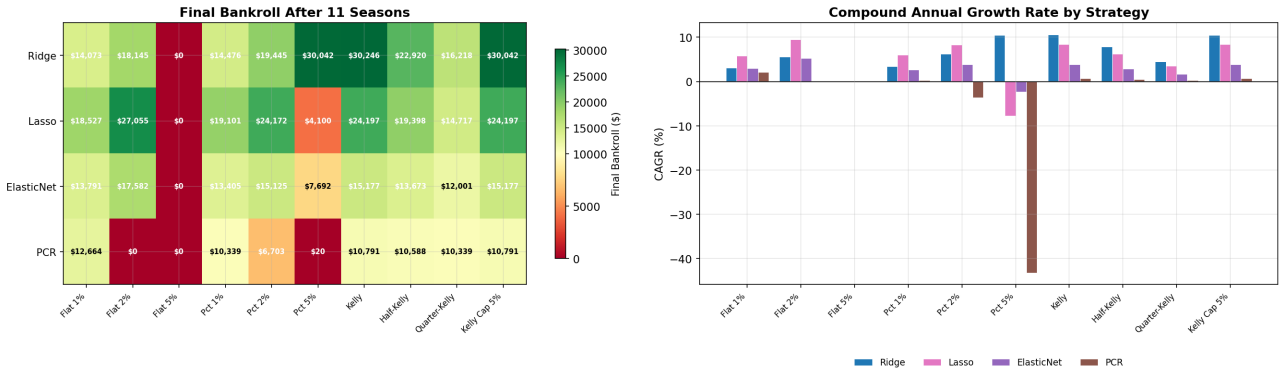

Figure 16: Backtest heatmap of final bankroll values and CAGR across all model-strategy combinations.

The simulations reveal that sizing matters as much as accuracy. Flat $500 bets cause bankruptcy for all four models despite positive edges. Full Kelly achieves the highest CAGR but with extreme drawdowns (76.6% for Ridge). Fractional Kelly variants consistently offer better risk-adjusted returns.

Figure 18: Risk-Return Frontier — Walk-Forward Backtest

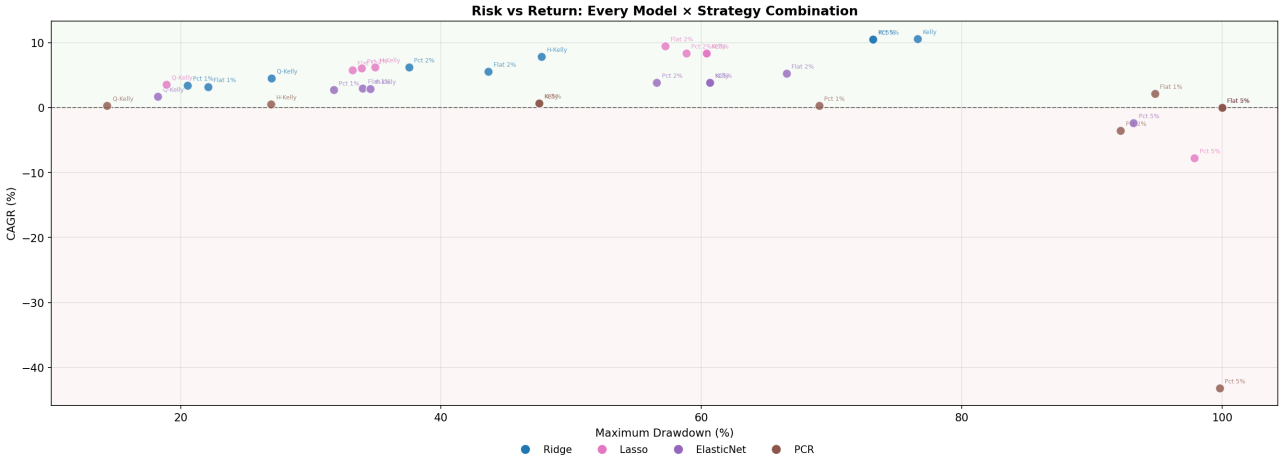

Figure 18: Risk-return scatter of all model-strategy combinations. The Pareto frontier runs through Ridge's conservative strategies.

Model Nomination

Using only backtest data available before the 2024/25 season, we nominate Lasso with Flat 2% sizing ($200 per bet) as the deployment strategy. While Ridge offers marginally higher CAGR (+10.6% vs. +9.5%), Lasso wins on every other criterion: lower maximum drawdown (57.2% vs. 76.6%), more winning seasons (7/11 vs. 6/11), 5.5 times more bets (4,529 vs. 826) for greater statistical reliability, and a simpler sizing rule that requires no win-probability estimation. This nomination is entirely prospective.

Holdout Results: 2024/25 Season

The Lasso model places 256 qualifying bets on the unseen 2024/25 season, achieving 53.1% accuracy. This is within 0.3 percentage points of its 11-season backtest accuracy of 53.4%, and sits above the 52.4% break-even threshold. Under Flat 2% sizing, the $10,000 bankroll grows to $10,727, a +7.3% return with a maximum drawdown of 15.3%.

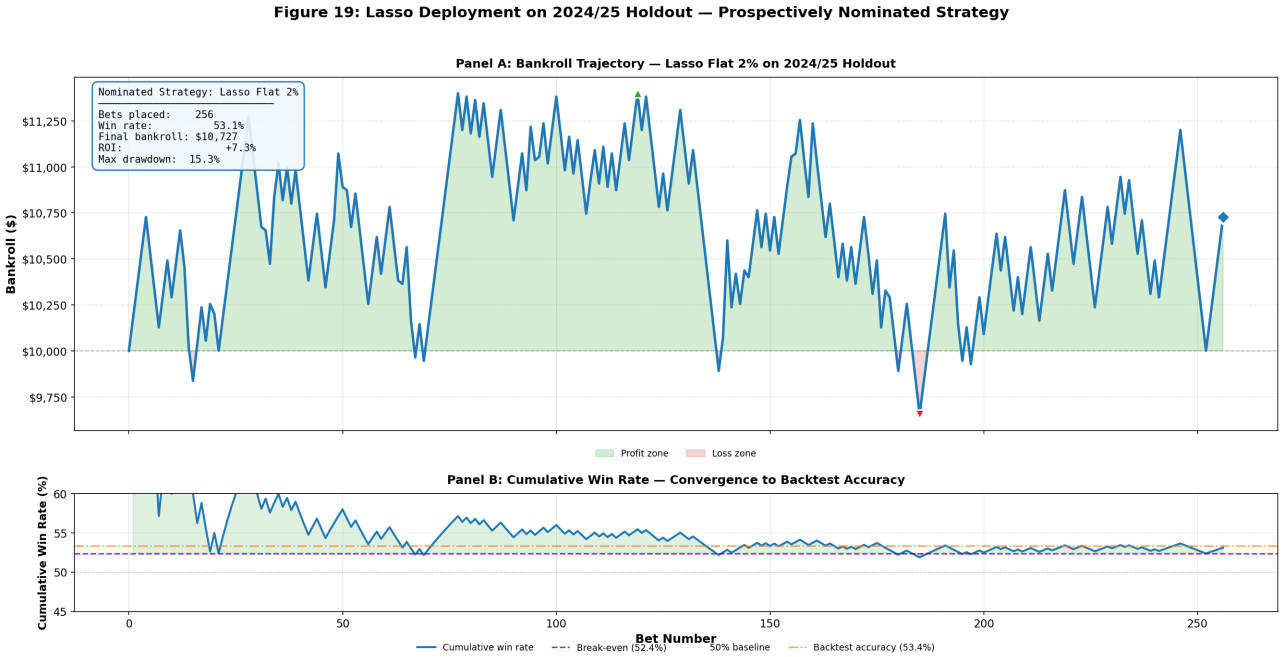

Figure 19: Lasso Flat 2% deployment on the 2024/25 holdout season — bankroll trajectory and cumulative win rate convergence.

The consistency between backtest and holdout is the most important finding: the strategy nominated prospectively on the basis of 11 seasons of data remains profitable on genuinely unseen data. As an investment benchmark, the strategy's +7.3% return compares to the S&P 500's +3.7% over the same October 2024 to June 2025 window, with near-zero correlation to equity markets.

A single season of 256 bets is statistically underpowered to prove a persistent edge. The 95% Wilson confidence interval for holdout accuracy spans [47.0%, 59.1%], which includes both 50% and 52.4%. The positive result is encouraging but not conclusive.

8. Real-World Implementation Constraints

Before drawing conclusions about practical viability, we examine three frictions that stand between a profitable backtest and actual profits: execution slippage, market depth, and account restrictions.

Execution Slippage

We simulate scenarios where the betting line shifts 0.5 to 3.0 points against the bet direction before execution. At zero slippage, Lasso achieves its full +1.9% flat-bet ROI on 4,529 backtest bets. The break-even slippage is approximately 0.5 points, which falls at the lower bound of typical NBA line movement (0.5-1.5 points). Even modest execution delays can eliminate the advantage entirely.

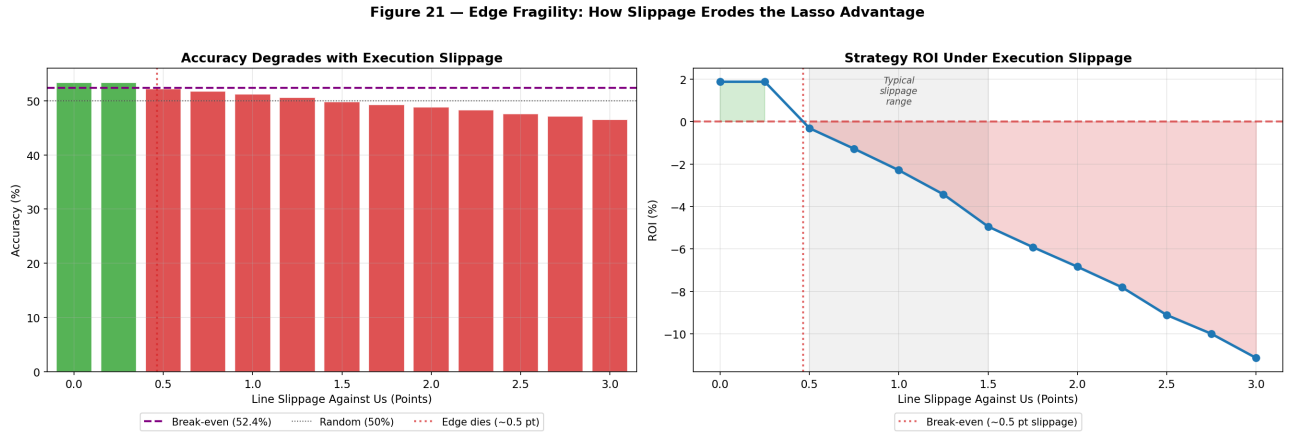

Figure 21: Accuracy and ROI degradation under increasing execution slippage. The typical slippage range (grey band) largely overlaps with the unprofitable region.

Market Depth and Scalability

Sportsbook betting limits cap the maximum bet per game at $2,500-$10,000 depending on the book. Across approximately 10 accessible US sportsbooks, the maximum deployable capital per game is roughly $50,000, implying a ceiling bankroll of $2.5 million at Flat 2% sizing. A $10,000 bankroll generates an expected annual profit of approximately $188; a $250,000 bankroll generates approximately $4,700. Scaling up capital does not fix the slippage problem, as break-even slippage is a property of the edge, not the bankroll size.

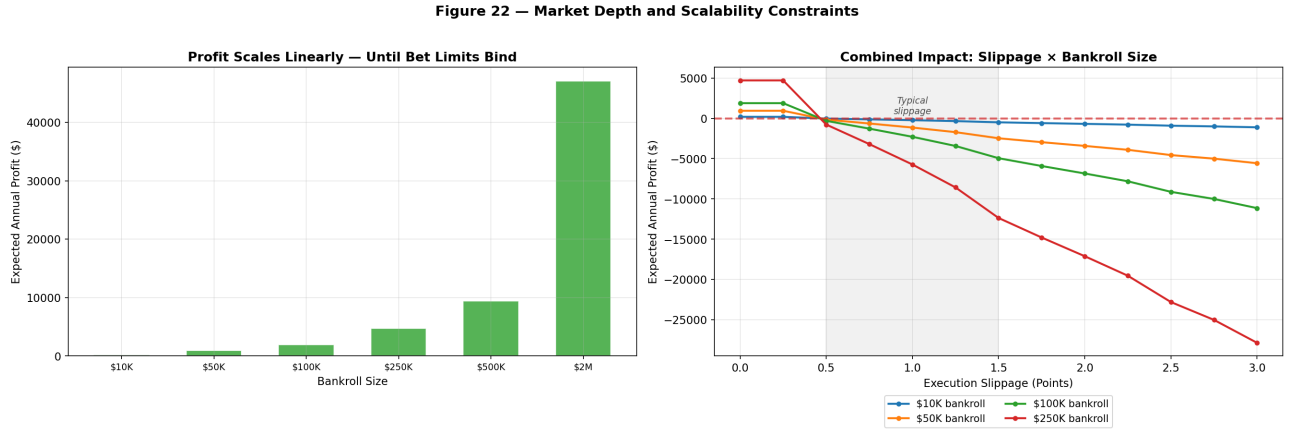

Figure 22: Expected profit by bankroll size and combined impact of slippage and capital deployment.

Account Restrictions

The most significant threat is account restrictions. Sportsbooks track betting patterns and flag accounts that exhibit Closing Line Value (consistently getting better odds than the final line). Winning accounts are typically limited or closed within 1-3 seasons. Our Lasso model places approximately 412 bets per season at a thin 53.4% edge, a volume and pattern that would likely trigger restrictions within 2-3 seasons.

9. Conclusion

This project set out to determine whether machine learning models can profitably predict NBA over/under totals. The answer is nuanced: a genuine statistical edge exists, but real-world frictions make it extremely difficult to capture reliably.

The Edge Is Real

Linear models (Ridge, Lasso, ElasticNet) achieve betting accuracies of 51.7–52.0% on over 11,000 out-of-sample predictions, with p-values below 0.001 after Bonferroni correction. Selective betting pushes four models above the 52.4% break-even threshold, with Ridge achieving 55.0% accuracy and +5.4% ROI on 826 high-confidence bets.

Simple Models Dominate

Simple linear models consistently outperform complex nonlinear alternatives (XGBoost, RBF+Ridge). XGBoost shows a 2.3-point train-vs-OOS RMSE gap, confirming that it memorizes training patterns that do not generalize. In an efficient market where exploitable signal is small and noisy, the additional flexibility of complex models becomes a liability rather than an asset. This is a textbook illustration of the bias-variance tradeoff in a low-signal environment.

The Holdout Confirms the Backtest

Our prospectively nominated Lasso Flat 2% strategy achieved 53.1% accuracy on 256 unseen 2024/25 bets, within 0.3 percentage points of its 11-season backtest accuracy of 53.4%. The $10,000 bankroll grew to $10,727 (+7.3% return), closely tracking the +9.5% annualized CAGR observed in the backtest. This consistency between prospective nomination and holdout performance is the most important result of the project, though a single season of 256 bets is insufficient for statistical certainty.

Real-World Viability Is Limited

The strategy passes three of six viability criteria (backtest accuracy, backtest ROI, holdout confirmation) but fails the remaining three. The break-even slippage of approximately 0.5 points falls at the lower bound of typical NBA line movement (0.5–1.5 points), meaning even modest execution delays can eliminate the edge. Sportsbook account restrictions track betting patterns and flag accounts that exhibit Closing Line Value, imposing a hard time limit of 1–3 seasons before the account is limited or closed.

Implications for Market Efficiency

Our results are consistent with the semi-strong form of market efficiency: publicly available statistical information is largely priced into the Vegas line. The residual edge (+1.9% flat-bet ROI in backtest, +7.3% bankroll ROI in the holdout season) is real but small enough that transaction costs and market structure constraints make it extremely difficult to capture reliably. From a machine learning perspective, this project demonstrates that model accuracy and economic viability are fundamentally different questions. A model can be statistically superior to random chance ($p < 0.001$) yet still fail to generate profit after accounting for the market's built-in margin. The gap between statistical significance and economic significance is the defining challenge of applied ML in efficient markets.